Mixture-of-Hierarchical Experts: Optimized Mamba Architecture for Vision Diffusion

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Abstract

Existing Mamba-based diffusion backbones primarily rely on sequential propagation, providing limited mechanisms for explicitly modeling spatial and multi-scale representations. We address this limitation by proposing **Mixture-of-Hierarchical Experts (MoH)**, a Mamba-based diffusion backbone that introduces hierarchical Mamba scanning across multiple spatial resolutions. This hierarchy allows Mamba to capture complementary representations ranging from coarse global structure to fine-grained spatial details. However, the representations and computations required for denoising vary substantially across timesteps, making a fixed hierarchical computation suboptimal. Rather than using a fixed hierarchy, MoH organizes choices in representation scale, feature transformation, and Mamba propagation depth into a structured expert space. These choices are dynamically coordinated according to the denoising state, allowing both spatial representation and sequential processing to evolve throughout denoising. At comparable model scales, MoH substantially improves generation quality in both unconditional and text-conditional settings, reducing FID from 14.27 to 7.99 on CelebA-HQ 256×256 and from 41.80 to 20.00 on MS-COCO, while also reducing GFLOPs and inference latency. Code is available at https://github.com/yejun2/Mixture-of-Hierarchical-Experts.

1 Introduction

Diffusion models have emerged as the dominant paradigm for image generation, achieving state-of-the-art performance across a wide range of vision tasks [20, 11, 38, 41, 45, 51]. Recent progress has been driven by advances in backbone architectures, with U-Net-based models [39, 38, 40] providing strong multi-scale inductive biases [9], and Transformer-based models offering powerful global context modeling [34, 2, 4, 31, 18, 6, 27, 32]. However, these approaches suffer from fundamental computational inefficiency [8, 10]. In particular, Transformer-based diffusion models rely on global self-attention with $\mathcal{O}(N^2)$ complexity, and in diffusion models, backbone computation is repeatedly applied across denoising steps, amplifying the overall computational cost.

To address this challenge, recent work has explored linear-time architectures such as State Space Models (SSMs) [48], notably Mamba [16], which enables sequence modeling with $\mathcal{O}(N)$ complexity and offers a promising direction for scalable diffusion backbones. However, directly applying Mamba to vision tasks remains challenging, as visual data is inherently two-dimensional, while Mamba operates through one-dimensional sequential propagation [52, 29, 22, 17, 35, 50]. This mismatch limits its ability to effectively capture spatial topology and multi-scale features.

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Figure 1: Samples generated by MoH-B on CelebA-HQ at 256 × 256 resolution.

Existing approaches, such as ZigMa [21], attempt to alleviate this issue through multi-directional or stochastic scanning strategies that improve spatial coverage. However, these methods remain fundamentally sequential and primarily modify the traversal order, rather than explicitly modeling how spatial information should be structured and aggregated. As a result, spatial features are still processed in a largely uniform manner, without a mechanism to adaptively integrate information across resolutions or spatial regions.

At the same time, diffusion models exhibit inherently timestep-dependent behavior. At high-noise timesteps, effective denoising relies on capturing coarse global structures, whereas at low-noise timesteps, the model must refine fine-grained details [7, 1]. Despite this, existing Mamba-based approaches apply identical computations across timesteps, lacking an explicit mechanism to adapt representation, computation depth, and feature mixing to the evolving requirements of the denoising process. Consequently, they often rely on uniformly deep architectures to handle all cases, leading to inefficient computation and suboptimal feature utilization.

In this work, we propose **Mixture-of-Hierarchical Experts (MoH)** to address these limitations. Unlike existing Mamba backbones that process visual tokens through sequential scanning at a fixed resolution, MoH organizes Mamba scanning hierarchically across multiple spatial resolutions, providing an explicit multi-scale spatial inductive bias. Rather than using this hierarchy as a fixed architecture, MoH formulates the architectural choices arising from hierarchical scanning as a structured expert space that can adapt to the current denoising state. Specifically, MoH adapts the active representation scales, the degree of spatial mixing before compression, and Mamba propagation depth. The latter provides a complementary control over representation, as deeper Mamba layers tend to form higher-level representations while shallower layers retain more localized information [23]. An integrated routing controller coordinates these architectural decisions to construct a computation path that evolves throughout denoising.

Contributions.

- We introduce a **hierarchical Mamba Architecture** that performs Mamba scanning across multiple spatial resolutions, providing explicit multi-scale spatial structure for visual generation beyond flat sequential processing.
- Rather than using hierarchy as a fixed architecture, we formulate the **hierarchy architectural choices**—including representation scale, feature transformation, and Mamba propagation depth—as a structured expert space that can be adaptively selected throughout denoising.
- We introduce **coordinated routing** in denoising states, where SSR, CMR, and EDR jointly adapt active representation scales, spatial mixing and compression, and Mamba propagation depth to construct an appropriate computation path for each denoising state.
- MoH substantially improves generation quality on both CelebA-HQ and MS-COCO while reducing computational cost and inference latency. As shown in Fig. 1, MoH also generates visually coherent samples with fine-grained details.

2 Related Work

Diffusion Backbones. Early diffusion models [20] adopted U-Net [39] as the standard denoising backbone, and it has remained the dominant choice in much of the subsequent literature. U-Net follows an encoder–decoder architecture that progressively compresses the input into hierarchical representations and reconstructs the output through a coarse-to-fine decoding process, with skip connections. This design endows the network with a strong multi-scale inductive bias, while the skip connections directly propagate high-resolution information from the encoder to the decoder. Such properties align naturally with the requirements of diffusion models, where the denoising network must repeatedly recover both global structure and local detail from noisy inputs at every timestep, making U-Net a particularly effective backbone for this task.

More recently, Transformer-based architectures [34, 2] have emerged as a competitive alternative and are now widely adopted in large-scale diffusion models [13]. Unlike U-Net backbones, these models discard the convolutional encoder–decoder hierarchy. Instead, the noisy latent is partitioned into non-overlapping patches, projected into tokens, and processed by a stack of homogeneous Transformer blocks. Timestep and conditioning information are injected via adaptive layer normalization. This design enables global receptive fields from the first layer through self-attention and removes the architectural asymmetry between encoder and decoder. It also exhibits favorable scaling behavior, as performance improves predictably with model size, data, and compute. These properties are particularly beneficial for diffusion models in high-dimensional latent spaces, where capturing long-range dependencies and scaling to large model sizes are critical. This explains the strong empirical performance of Transformer backbones in modern text-to-image systems.

More recently, State Space Models (SSMs), particularly Mamba [16], have been adapted to 2D vision tasks [52, 29] and have been explored as efficient alternatives to attention-based backbones for diffusion [47, 49, 21]. Mamba treats the input as a one-dimensional sequence and propagates information through a selective state-space update, where transition and projection parameters are conditioned on the input. The full sequence is processed in linear time. This design provides linear complexity in sequence length and introduces input-adaptive global memory, making it appealing for high-resolution latents whose token counts grow rapidly with image size. However, existing Mamba-based diffusion models still fall short of Transformer-based counterparts in generation quality. The underlying 1D causal scan imposes an arbitrary ordering on inherently 2D latents, leading to a range of hand-designed scan strategies, including bidirectional [52], cross-directional [29], and zigzag [21] scans. While these approaches improve spatial coverage, they introduce biases toward specific traversal patterns, with no superior strategy. The compressed recurrent state limits dense long-range interactions that self-attention captures more directly. As a result, designing competitive and scan-agnostic SSM-based diffusion backbones remains an open problem.

Adaptive Computation and Routing. A complementary line of research investigates adaptive computation [43, 15, 37, 14]. This direction is motivated by the observation that not all inputs require the same amount of compute, and that conditionally activated subnetworks can improve the capacity–efficiency trade-off over uniformly dense models. These methods introduce learned routing mechanisms that selectively activate a subset of parameters per input. For example, mixture-of-experts layers dispatch each token to a small number of expert sub-networks via a gating function, while dynamic-depth and early-exit models adjust the number of executed layers based on input difficulty. Such conditional execution allows total model capacity to scale with the number of experts or layers without a proportional increase in per-input compute. It also enables different inputs to specialize in different parameter subsets, leading to improved efficiency and performance across language and vision tasks. However, existing methods primarily perform routing at the token or layer level within a fixed architecture. As a result, higher-level design choices—such as the allocation of computation across feature scales, network depth, and feature compression—remain uniform across inputs. In contrast, our approach performs routing at the architectural level, enabling timestep-conditioned adaptation along these dimensions.

3 Preliminaries

Diffusion Models. We consider a standard forward noising process at timestep t:

$$x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{1}$$

and train a network f_θ to predict a target (e.g., velocity v) [25]:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon}\left[\|f_\theta(x_t, t, c_t) - v\|^2\right], \tag{2}$$

Here, c_t denotes the conditioning signal, including timestep information and optional inputs such as text embeddings. During inference, samples are generated by iteratively applying the learned denoising function.

Mamba. Mamba [16] is a selective state space model that updates hidden states sequentially [17]:

$$h_i = A_i h_{i-1} + B_i x_i, \quad y_i = C_i h_i, \tag{3}$$

where (A_i, B_i, C_i) are input-dependent parameters. In practice, Mamba blocks are used with residual connections, preserving input information while enabling progressive feature transformation. Stacking multiple blocks increases the effective receptive field, allowing deeper configurations to capture stronger global dependencies, whereas shallower configurations preserve more local structure.

4 Mixture-of-Hierarchical Experts

4.1 Overview

We propose *Mixture-of-Hierarchical Experts (MoH)*, a Mamba-based diffusion backbone that organizes Mamba computation into a spatial hierarchy. As illustrated in Fig. 2, the encoder performs Mamba scanning across multiple resolutions, producing hierarchical representations with progressively different spatial scales.

MoH interprets the architectural choices created by this hierarchy as a structured expert space. Given the current denoising state, SSR selects the active hierarchical stages, while CMR and EDR adapt feature transformation and Mamba propagation depth within the selected stages. These routing decisions are coordinated by the integrated routing controller (IRC), enabling complementary adaptation across routing spaces for the current denoising state.

The lowest active stage serves as a dynamic bottleneck. From this bottleneck, the decoder progressively upsamples and fuses the routed encoder representations to reconstruct the output in a coarse-to-fine manner. The following sections describe the hierarchical Mamba architecture, routing mechanism, routing modules, and prediction head in detail.

4.2 Hierarchical Mamba Architecture

MoH employs an S-stage hierarchical encoder that progressively forwards features across multiple spatial resolutions. Given the input feature z_s at stage s, EDR first determines the stage-wise Mamba computation and produces the processed representation h_s. CMR then transforms and spatially compresses h_s to construct the input z_{s+1} for the next stage:

$$z_{s+1} = \mathrm{CMR}_s\left(h_s, c_t, \alpha_{t,s}^{\mathrm{CMR}}\right), \quad h_s = \mathrm{EDR}_s\left(z_s, c_t, \alpha_{t,s}^{\mathrm{EDR}}\right), \qquad s = 1, \ldots, S-1. \tag{4}$$

Here, c_t denotes the conditioning signal at timestep t, incorporating the timestep embedding and optional conditions such as text embeddings, while $\alpha_{t,s}^{\mathrm{EDR}}$ and $\alpha_{t,s}^{\mathrm{CMR}}$ denote the corresponding routing decisions. Importantly, the stages do not operate as independent multi-scale branches. Instead, they form a sequential hierarchical forwarding path: each stage processes the representation received from the previous scale, and its routed output is subsequently transformed and forwarded to the next, lower-resolution stage. Consequently, the representation constructed at each stage also determines the information propagated to deeper stages.

As features are forwarded through the hierarchy, their spatial resolution progressively decreases. By scanning at multiple resolutions, the hierarchy provides explicit multi-scale spatial structure and allows each stage to use relatively shallow Mamba stacks, since lower-resolution stages can capture broader spatial context without relying solely on additional Mamba depth. This progressive forwarding enables to capture complementary representations from global semantics to fine details.

For text-conditioned generation, textual information is incorporated throughout this hierarchical forwarding process. In the encoder, text embeddings are injected through cross-attention during stage-wise feature processing, allowing the intermediate representations at different resolutions to be

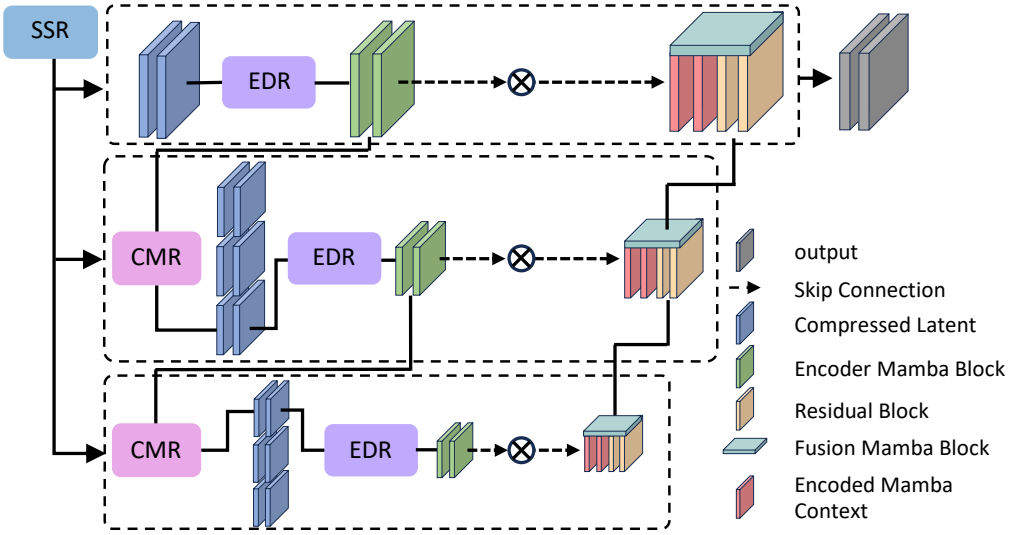


Figure 2: Overview of Mixture-of-Hierarchical Experts (MoH). MoH treats a hierarchical backbone as a structured expert space and dynamically constructs a timestep-dependent computation path via routing over architectural decisions.

conditioned on the input text. In the decoder, stage-wise projections of the text embedding are used during coarse-to-fine reconstruction, so that textual conditioning is preserved as hierarchical features are progressively integrated across resolutions.

4.3 Routing Mechanism: Integrated Routing Controller (IRC)

As shown in Fig. 3, we employ a unified routing framework that adapts hierarchical computation to the current denoising state. Given timestep t and conditioning signal y, we construct a shared conditioning representation c_t and extract stage-wise statistics ϕ_t from feature F_t of input. The IRC additionally predicts a global scale prior p_t, which provides shared routing context across the three routing modules. Without IRC, each router is optimized independently without this shared routing prior, whereas IRC couples their decisions through p_t to encourage coordinated optimization across routing spaces. Based on these signals, SSR, CMR, and EDR make specialized decisions for stage selection, compression and mixing, and Mamba depth, respectively.

4.4 Routing Modules

Stage Selection. SSR selects a subset of hierarchical stages via adaptive sparse routing. Given routing logits b_t^{SSR}, we compute activation probabilities and select stages using a threshold τ:

$$\rho_{t,r} = \text{sigmoid}(b_{t,r}^{\text{SSR}}), \quad r \in \mathcal{S}, \qquad m_{t,r}^{(0)} = \mathbf{1}[\rho_{t,r} > \tau], \quad k_t^{(0)} = \sum_r m_{t,r}^{(0)}. \tag{5}$$

where $\mathcal{S}$ denotes the set of candidate stage resolutions. To prevent degenerate sparsity, we enforce a minimum number $k_{\min}$ of active stages:

$$m_{t,r} = \begin{cases} m_{t,r}^{(0)}, & k_t^{(0)} \geq k_{\min}, \\ \mathbf{1}[r \in \text{Top-K}_{k_{\min}}(b_t^{\text{SSR}})], & k_t^{(0)} < k_{\min}. \end{cases} \tag{6}$$

We normalize the hard selection and activation probabilities and apply a straight-through estimator [3] to obtain the final stage-selection weights:

$$\tilde{m}_{t,r} = \frac{m_{t,r}}{\sum_j m_{t,j}}, \qquad \tilde{\rho}_{t,r} = \frac{\rho_{t,r}}{\sum_j \rho_{t,j}}, \qquad \alpha_{t,r}^{\text{SSR}} = \tilde{m}_{t,r} + \tilde{\rho}_{t,r} - \text{sg}(\tilde{\rho}_{t,r}). \tag{7}$$

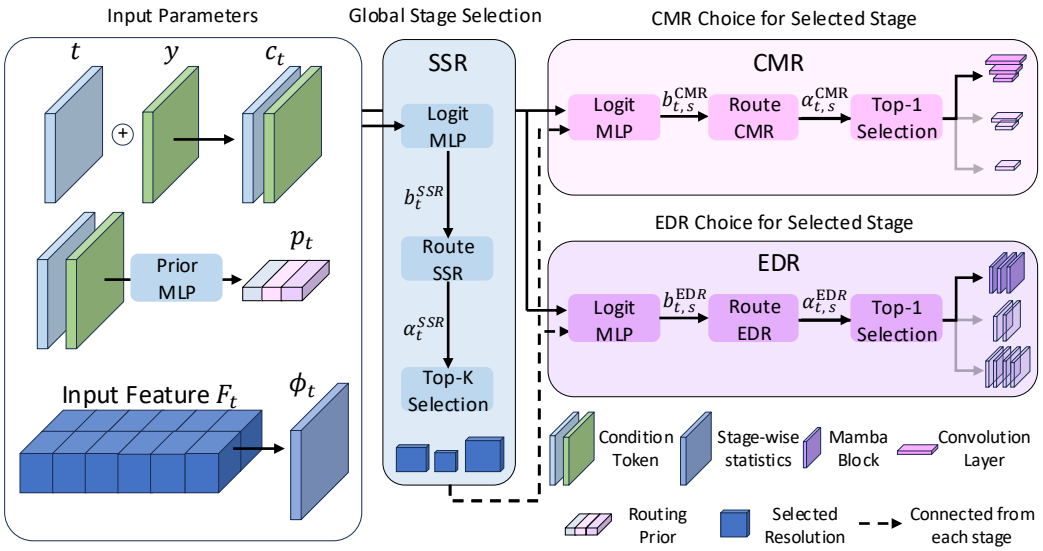


Figure 3: Integrated routing mechanism of MoH. A shared conditioning signal and routing prior coordinate SSR, CMR, and EDR to adapt stage selection, feature mixing, and Mamba propagation depth.

The selected stage set $\mathcal{S}_t = \{r \in \mathcal{S} \mid \alpha_{t,r}^{\mathrm{SSR}} > 0\}$ defines the active representation levels, and the deepest stage determines the bottleneck resolution. This dynamic bottleneck enables adaptive control over the level of abstraction, using deeper stages for global structure modeling and shallower stages for fine-grained denoising.

Feature Mixing and Compression. CMR selects a transformation branch via Top-1 routing:

$$\rho_{t,s}^{\mathrm{CMR}} = \mathrm{softmax}\big(b_{t,s}^{\mathrm{CMR}}\big)\,, \qquad \alpha_{t,s}^{\mathrm{CMR}} = \text{Top-1}\big(\rho_{t,s}^{\mathrm{CMR}}\big)\,. \tag{8}$$

where $b_{t,s}^{\mathrm{CMR}}$ denotes the routing logits. CMR is trained using the same straight-through formulation as Eq. 7. We define three transformation presets, *weak*, *base*, and *strong*, corresponding to convolutional depths of 0, 1, and 2, respectively, before a shared stride-2 downsampling operation. These presets control the amount of local spatial processing applied before Mamba operates on the compressed representation. Consequently, CMR provides a CNN-like spatial inductive bias whose strength can be adapted to the current denoising state, rather than relying solely on a fixed sequential scan for spatial interaction. CMR is applied when constructing the 16×16 and 8×8 hierarchical representations.

Mamba Scanning Depth. EDR selects the number of Mamba blocks via Top-1 routing:

$$\rho_{t,s}^{\mathrm{EDR}} = \mathrm{softmax}\big(b_{t,s}^{\mathrm{EDR}}\big)\,, \qquad \alpha_{t,s}^{\mathrm{EDR}} = \text{Top-1}\big(\rho_{t,s}^{\mathrm{EDR}}\big)\,. \tag{9}$$

where $b_{t,s}^{\mathrm{EDR}}$ denotes the routing logits. EDR is trained using the same straight-through formulation as Eq. 7. We define *shallow*, *base*, and *deep* configurations using shared-prefix Mamba blocks. Deeper routes extend the same shared prefix with additional Mamba blocks according to the selected preset; detailed stage-wise configurations are provided in Table 8.

Each additional Mamba block applies another input-dependent selective state transition, so the selected depth directly controls how information is repeatedly propagated and aggregated along the scan path. Previous observations in Vision Mamba [23] further suggest that deeper layers tend to form higher-level representations, whereas shallower layers retain more localized information. EDR therefore treats Mamba propagation depth as a denoising-state-dependent control axis, rather than merely varying generic network capacity.

Discussion. This asymmetric routing design reflects the distinct functional roles of each module, enabling complementary adaptation across representation, feature transformation, and computational depth. Together, these coordinated routing mechanisms allow the model to jointly adapt representation and computation across timesteps, resulting in a more efficient and expressive denoising process.

4.5 Prediction Head

The decoder reconstructs the output by progressively integrating hierarchical representations in a coarse-to-fine manner. Starting from the bottleneck representation, it iteratively upsamples and refines features at each stage. At each stage, the upsampled decoder feature is fused with the corresponding routed encoder feature via gated summation:

$$\hat{u}_{s-1} = \mathrm{Up}(u_s) + G_{s-1} \odot \tilde{z}_{s-1}, \qquad u_{s-1} = M_{s-1}(\hat{u}_{s-1}, c_t), \tag{10}$$

where stage s denotes a lower-resolution level and $s-1$ a higher-resolution level. Here, u_s denotes the decoder feature at stage s, and $\hat{u}_{s-1}$ is the fused feature obtained by combining the upsampled decoder feature with the routed encoder feature at stage $s-1$. $\tilde{z}_{s-1}$ is the routed encoder feature, G_{s-1} is a learnable gating function, and M_{s-1} is a Mamba block for feature refinement.

The fusion combines low-resolution semantic context from the decoder with higher-resolution gated encoder features, while the Mamba block refines the fused representation by integrating global context and local details. Importantly, the encoder features $\tilde{z}_{s-1}$ are not raw activations but outputs of hierarchical processing via EDR, which already incorporate Mamba-based transformations. Thus, the decoder operates on multi-level representations encoding both semantic structure and fine-grained details. Under conditioning, both encoder and decoder features are modulated by the shared signal c_t, and the refinement step further integrates this information. This enables the reconstruction process to adapt both feature integration and refinement based on the input condition.

5 Experiments

5.1 Datasets and Training Details

Datasets. We conduct experiments on CelebA-HQ 256×256 [24] and MS-COCO 256×256 [28]. For CelebA-HQ, we use precomputed latent representations, where images are encoded into a $32 \times 32 \times 4$ latent space using the Stable Diffusion v1.5 VAE [26, 38]. For MS-COCO, we use the 2014 split with text conditioning. Each sample consists of a latent image representation ($32 \times 32 \times 4$) and a CLIP text embedding [36] of dimension 768.

Experiment Settings. We adopt velocity prediction [25] with timesteps sampled from $t \sim \mathcal{U}[0, 1]$. Models are trained with AdamW [30] using mixed precision and EMA (decay $= 0.9999$). We use 50-step sampling for validation and report final FID [19] using a 250-step ODE sampler (Dopri5) [44, 5]. All ablations are conducted consistently with AMP-FP16, while the final models use AMP-BF16 for improved numerical stability. We explicitly report these settings since prior Mamba baselines specify only mixed-precision training. Batch sizes are 64 for CelebA-HQ and 256 for MS-COCO. Additional details are provided in Appendix A.1.

Evaluation. We report Fréchet Inception Distance (FID) as the primary evaluation metric [19]. We additionally report IS [42], sFID [33], and FDD [46] for comprehensive evaluation. All FID results are computed using the EMA model.

5.2 Results on Image Generation

CelebA-HQ 256×256**.** As shown in Table 1, all models other than ZigMa are trained for 100K steps under identical settings, while the ZigMa result is taken from the original paper. MoH-B achieves an FID of 7.99, substantially improving over the 14.27 FID of ZigMa [21], while using 129M parameters and 80.88G FLOPs, compared with 133M and 97.52G for ZigMa. MoH also consistently outperforms both SSM-based GSPN [48] and Transformer-based SiT [32] across the compared model scales. These results demonstrate that hierarchical scanning Mamba architecture with timestep-conditioned routing substantially improves generation quality without relying on increased model scale or computation.

Table 1: Generation quality and model complexity of diffusion backbones on CelebA-HQ 256×256. Except for ZigMa, all models are trained for 100K steps under identical training settings. The ZigMa FID is taken from the original paper.

Method	Backbone	FID^{5k} ↓	FLOPs (G) ↓	Params (M) ↓
GSPN-B/2 [48]	SSM	21.00	36.17	141
GSPN-L/2	SSM	20.71	118.14	461
SiT-B/2 [32]	Transformer	17.30	**23.02**	129
SiT-L/2	Transformer	15.65	80.74	456
ZigMa [21]	Mamba	14.27	97.52	133
MoH-S (ours)	Mamba	9.16	35.79	**57**
MoH-B (ours)	Mamba	**7.99**	80.88	129

Table 2: Comparison of Mamba-based diffusion backbones on text-conditional image generation on MS-COCO 256×256. All models follow a 400K-step training schedule.

Method	Venue	FID^{5k} ↓
VisionMamba [52]	ICML'24	60.20
ZigMa [21]	ECCV'24	41.80
U-shape Mamba [12]	CVPRW'25	39.10
MoH-B (ours)	–	**20.00**

Table 3: Inference efficiency comparison across models. **Thr.** denotes throughput (samples per second), **Lat.** denotes mean inference latency per sample (milliseconds), and **Mem.** denotes peak GPU memory usage (MiB).

Model	Thr. ↑	Lat. ↓	Mem. ↓
ZigMa [21]	51.170	19.543	573.32
MoH-B	**66.244**	**15.096**	**545.12**

MS-COCO 256×256**.** We evaluate MoH on the more complex MS-COCO dataset. As shown in Table 2, MoH-B achieves an FID of 20.00, substantially outperforming prior Mamba-based models, with 60.20 for VisionMamba [52], 41.80 for ZigMa [21], and 39.10 for U-shape Mamba [12]. Unlike MoH, U-Shape Mamba [12] employs a U-shaped encoder–decoder hierarchy over Mamba features, rather than treating multiple spatial resolutions as Mamba scanning spaces. The baseline results are adopted from the comparisons reported in the corresponding prior work. For a controlled comparison, MoH-B follows the same model scale as ZigMa; more specific comparison of each model is provided in Appendix A.2. These results demonstrate the effectiveness of timestep-conditioned hierarchical routing for complex, multi-object generation, which requires capturing both global semantics and fine-grained details.

Efficiency. We further evaluate the inference efficiency of MoH-B against Zigma, where Zigma refers to the ZigMa configuration reported in Table 1. All measurements are conducted with the CelebA-HQ generation checkpoints at batch size 1. As shown in Table 3, MoH-B improves throughput from 51.170 to 66.244 samples per second while reducing the mean inference latency from 19.543 ms to 15.096 ms per sample. This corresponds to a 29.5% increase in throughput and a 22.8% reduction in inference latency. MoH-B also reduces peak GPU memory usage from 573.32 MiB to 545.12 MiB. Together with the improvements in generation quality and model complexity reported in Table 1, these results demonstrate that MoH achieves more efficient inference without relying on increased model scale or computation.

5.3 Ablation Study

Hierarchical Architecture. We evaluate the effect of hierarchical modeling and stage configuration. As shown in Table 5, introducing a two-stage hierarchy $(32, 16)$ without routing reduces FID from 14.27 (ZigMa) to 13.17. Adding the 8×8 stage further reduces FID to 11.93, demonstrating the benefit of hierarchical modeling even without adaptive routing. However, extending the hierarchy to $(32, 16, 8, 4)$ worsens FID to 12.64, indicating that increasing hierarchy depth is not uniformly beneficial. We therefore adopt the three-stage configuration $(32, 16, 8)$, which achieves the lowest FID among the evaluated configurations, and exclude the 4×4 stage from the routing candidate set.

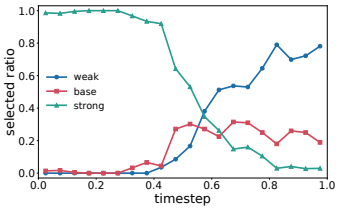


Figure 4: CMR behavior at stage 2 (16×16 resolution).

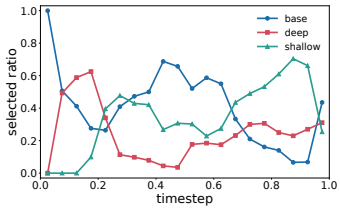


Figure 5: EDR behavior at stage 2 (16×16 resolution).

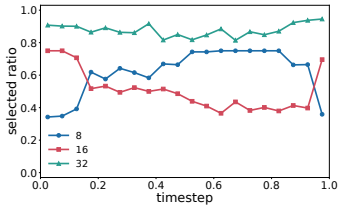


Figure 6: SSR behavior across timesteps.

Table 4: Scan strategy ablation on CelebA-HQ 256×256, varying only the scan in a fixed $(32, 16, 8)$ hierarchy.

Model	Scan type	FID^{5k} ↓
Hierarchy-only	zigzagN8 [21]	12.37
Hierarchy-only	v2 [52]	**11.93**

Table 5: Ablation of hierarchical stage configurations on CelebA-HQ 256×256.

Model	Stages	FID^{5k} ↓
ZigMa [21]	–	14.27
Hierarchy-only	$(32, 16)$	13.17
	$(\mathbf{32}, \mathbf{16}, \mathbf{8})$	**11.93**
	$(32, 16, 8, 4)$	12.64

Scan Strategy. We evaluate the sensitivity to scanning patterns in the hierarchical architecture. As shown in Table 4, different scanning strategies yield similar performance (12.37 vs. 11.93), indicating that the proposed formulation substantially reduces scan-order bias. This contrasts with prior scanning strategy-based approaches, where performance is highly sensitive to traversal design [52, 29, 22, 21].

Routing Components. As shown in Table 6, CMR and EDR independently improve the fixed hierarchical baseline, reducing FID from 11.93 to 10.60 and 10.81, respectively. However, simply combining the two routers without IRC does not preserve these gains, yielding a worse FID of 11.00. This indicates that increasing routing capacity alone is insufficient, since spatial mixing and Mamba propagation depth jointly determine the resulting computation path and therefore require coordinated decisions. Introducing IRC substantially improves the joint configuration from 11.00 to 9.51. Further enabling adaptive-k SSR reduces FID to 9.28, showing an additional benefit from dynamically selecting the active hierarchical stages. Note that the 9.28 FID in Table 6 uses the same MoH-B architecture reported in Table 1; the ablation result is obtained with AMP-FP16, while the final 7.99 result uses AMP-BF16.

Coordination of Routing Components. We define coordination as complementary routing between CMR and EDR, rather than independent optimization of their individual decisions. This dependency naturally follows the hierarchical computation order: CMR first determines how features are spatially mixed and compressed, and EDR subsequently determines the Mamba propagation depth for the resulting representation. Therefore, an effective EDR decision should account for the spatial configuration produced by CMR.

This distinction is reflected in the learned routing behavior. Without IRC, EDR exhibits routing patterns similar to the independently trained EDR-only policy and remains largely insensitive to the preceding CMR choice. With IRC, in contrast, the EDR distribution changes according to the selected CMR configuration, indicating that propagation depth is adapted complementarily to the spatial processing already performed by CMR. This behavioral difference is consistent with the ablation in Table 6: jointly enabling CMR and EDR without coordination does not preserve their individual gains, whereas introducing IRC improves FID from 11.00 to 9.51.

Routing Behaviors. The learned policies also exhibit a structured evolution over the denoising trajectory. As shown in Figs. 4–6, CMR progressively shifts toward weaker spatial mixing, EDR adapts its propagation depth, and SSR utilizes multiple hierarchical representations without collapsing to a single scale. Together, these behaviors are consistent with the coarse-to-fine nature of diffusion, with

Table 6: Ablation study of routing components on CelebA-HQ 256×256. All routing ablations use the $(32, 16, 8)$ hierarchy, which achieves the best FID^{5k} among the evaluated stage configurations. Each module provides complementary gains, and the integrated controller achieves the best FID^{5k} $\downarrow$ performance.

Model	SSR	CMR	EDR	IRC	FID^{5k} $\downarrow$	sFID^{5k} $\downarrow$	FDD^{5k} $\downarrow$	IS^{5k} $\uparrow$
MoH-B	top-3	–	–	–	11.93	10.56	3.45	3.04
MoH-B	top-3	✓	–	–	10.60	9.16	2.70	2.95
MoH-B	top-3	–	✓	–	10.81	8.93	2.93	2.80
MoH-B	top-3	✓	✓	–	11.00	9.22	3.14	2.86
MoH-B	top-3	✓	✓	✓	9.51	8.39	2.46	2.91
MoH-B (full)	adaptive-k	✓	✓	✓	**9.28**	8.50	2.74	2.95

Table 7: Conditional EDR routing under CMR decisions during early denoising (timestep $t \in [0, 0.33]$).

Model	CMR	EDR decision (%)		
		shallow	base	deep
w/o IRC	weak	0.0	0.0	**100.0**
	base	0.0	0.0	**100.0**
	strong	0.0	0.0	**100.0**
with IRC	weak	24.8	**74.5**	0.7
	base	12.4	33.3	**54.3**
	strong	38.7	14.0	**47.3**

the architectural computation path evolving as denoising progresses. Additional routing behaviors across timesteps and hierarchical stages are analyzed in Appendix A.4.

6 Conclusion

We present **Mixture-of-Hierarchical Experts (MoH)**, a Mamba-based diffusion backbone that improves both generation quality and efficiency through timestep-conditioned hierarchical routing. By treating hierarchical representations as a structured expert space, MoH dynamically selects feature levels, transformation strength, and Mamba depth to match the evolving requirements of the denoising process. MoH achieves consistent improvements over both Transformer- and Mamba-based baselines [34, 21], with particularly large gains on complex datasets such as MS-COCO [28]. These results demonstrate that coordinated adaptation across representation and computation is crucial for modeling complex generative processes. More broadly, our findings suggest that the limitations of Mamba in vision stem from the lack of adaptive mechanisms rather than the architecture itself. By introducing architectural-level routing, MoH provides a flexible and scalable framework, highlighting *adaptive architectural routing* as a promising direction for generative modeling.

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A Technical appendices and supplementary material

A.1 Implementation Details

We provide specific training details of MoH in Table 8.

Detailed Training Setup. We employ mixed-precision training with both AMP-FP16 and AMP-BF16. All ablation studies are conducted using AMP-FP16 for fair comparison, while final models are trained with AMP-BF16 for improved stability and performance. We verify that relative improvements are consistent across both settings. We use an exponential moving average (EMA) of parameters with decay 0.9999, and report results using the EMA model. For CelebA-HQ, we use a batch size of 16 per GPU (4 A6000 GPUs, total batch size 64). For MS-COCO, we use a batch size of 64 per GPU (4 A100 GPUs, total batch size 256).

Table 8: Implementation details of MoH-B.

Category	Configuration
Input	Latent size 32×32, channels 4, image resolution 256
Patch	patch size 1
Hierarchy	Encoder stages $\{32, 16, 8\}$
Stage Depth	High-res 4, hierarchy 3, bottleneck 4
Routing (SSR)	Adaptive-k stage routing over $\{32, 16, 8\}$; threshold 0.5, min stages 2, max stages 3; batch balancing enabled
Dynamic Bottleneck	Enabled; candidates $\{32, 16, 8\}$; decoder starts from the lowest selected resolution
Routing (CMR)	Top-1 compression routing at output stages $\{16, 8\}$; choices are premix depths $\{weak : 0, base : 1, strong : 2\}$ before stride-2 downsampling
Routing (EDR)	Top-1 encoder-depth routing at stages $\{32, 16, 8\}$; choices are shallow/base/deep depths: $32 : \{3, 4, 5\}$, $16 : \{2, 3, 4\}$, $8 : \{3, 4, 5\}$
Training Batch	Batch 64 (16×4) for CelebA-HQ 256×256 and Batch 256 (64×4) for MS-COCO 256×256
Mamba	Scan type v2; EDR uses shared-prefix depth routing

A.2 Model size specification

Table 9 summarizes the model scales used for comparison. On CelebA-HQ, MoH-S and MoH-B contain 57M and 129M parameters, respectively, while ZigMa contains 133M parameters. For text-conditioned MS-COCO generation, both ZigMa and MoH additionally employ cross-attention modules for text conditioning, resulting in larger parameter counts even under the same backbone scale. Accordingly, MoH-S and MoH-B contain 162M and 237M parameters on MS-COCO, compared with 214M for ZigMa.

Table 9: Model size comparison on CelebA-HQ and MS-COCO 256×256.

Dataset	Model	Params (M)	FID^{5k} $\downarrow$
CelebA-HQ 256×256	ZigMa	133	14.27
	MoH-S	57	9.16
	MoH-B	129	7.99
MS-COCO 256×256	ZigMa	214	41.80
	MoH-S	162	22.32
	MoH-B	237	20.00

A.3 Sampling and Evaluation Details

We use the adaptive Dormand–Prince (Dopri5) solver for sampling with $\mathrm{atol} = 10^{-6}$ and $\mathrm{rtol} = 10^{-3}$. The initial latent is sampled from a standard Gaussian distribution with shape $4 \times 32 \times 32$, and evaluation is performed using EMA model parameters. For MS-COCO, classifier-free guidance is not used ($\mathrm{CFG} = 1.0$).

During online FID validation, we request 50 output time points from the ODE solver, while 250 output time points are used for offline sampling. Since Dopri5 uses adaptive integration, these values specify the sampled output trajectory rather than a fixed number of solver steps.

A.4 Routing Coordination Analysis and Routing Behavior

Table 10: Timestep-wise CMR routing distributions (%).

Timestep	CMR-only			CMR+EDR w/o IRC			CMR+EDR with IRC		
	weak	base	strong	weak	base	strong	weak	base	strong
0.05	50.0	0.0	50.0	50.0	50.0	0.0	0.0	51.6	48.4
0.15	0.0	50.0	50.0	61.0	38.6	0.4	5.3	39.8	54.9
0.25	0.0	50.0	50.0	39.7	50.5	9.8	53.0	7.0	39.9
0.35	41.9	50.0	8.1	1.3	48.7	50.0	89.2	1.4	9.5
0.45	50.0	47.0	3.0	0.0	50.0	50.0	96.1	0.2	3.7
0.55	58.4	0.0	41.6	0.0	50.0	50.0	98.5	0.0	1.5
0.65	77.1	0.0	22.9	47.9	2.1	50.0	99.3	0.0	0.7
0.75	96.3	0.0	3.7	49.8	0.2	50.0	99.4	0.0	0.6
0.85	100.0	0.0	0.0	50.0	0.0	50.0	100.0	0.0	0.0
0.95	100.0	0.0	0.0	50.0	0.0	50.0	100.0	0.0	0.0

Table 11: Timestep-wise EDR routing distributions (%).

Timestep	EDR-only			CMR+EDR w/o IRC			CMR+EDR with IRC		
	shallow	base	deep	shallow	base	deep	shallow	base	deep
0.05	0.0	33.3	66.7	0.0	0.0	100.0	33.3	1.4	65.4
0.15	0.0	33.3	66.7	0.0	0.0	100.0	34.8	51.1	14.1
0.25	33.3	66.7	0.0	0.0	0.0	100.0	36.3	62.0	1.7
0.35	33.3	66.7	0.0	0.0	66.7	33.3	34.0	65.7	0.3
0.45	66.7	33.3	0.0	0.0	66.7	33.3	27.9	71.9	0.2
0.55	100.0	0.0	0.0	0.0	100.0	0.0	22.9	77.0	0.1
0.65	100.0	0.0	0.0	33.3	33.3	33.3	11.3	88.4	0.3
0.75	100.0	0.0	0.0	33.3	33.3	33.3	0.6	95.2	4.2
0.85	100.0	0.0	0.0	100.0	0.0	0.0	0.0	99.2	0.8
0.95	100.0	0.0	0.0	100.0	0.0	0.0	0.7	99.2	0.2

Table 12: Conditional EDR routing distributions under CMR decisions, comparing models with and without IRC.

Timestep range	Model	CMR decision	EDR decision (shallow / base / deep)
Early (0.00–0.33)	w/o IRC	weak ($n = 1545$, 50.3%)	0.0% / 0.0% / 100.0%
		base ($n = 1424$, 46.4%)	0.0% / 0.0% / 100.0%
		strong ($n = 103$, 3.4%)	0.0% / 0.0% / 100.0%
	with IRC	weak ($n = 597$, 19.4%)	24.8% / 74.5% / 0.7%
		base ($n = 1008$, 32.8%)	12.4% / 33.3% / 54.3%
		strong ($n = 1467$, 47.8%)	38.7% / 14.0% / 47.3%
Mid (0.34–0.66)	w/o IRC	weak ($n = 504$, 12.3%)	0.0% / 0.0% / 100.0%
		base ($n = 1544$, 37.7%)	0.0% / 33.2% / 66.8%
		strong ($n = 2048$, 50.0%)	0.0% / 100.0% / 0.0%
	with IRC	weak ($n = 3923$, 95.8%)	33.2% / 66.5% / 0.3%
		base ($n = 16$, 0.4%)	12.5% / 87.5% / 0.0%
		strong ($n = 157$, 3.8%)	99.4% / 0.6% / 0.0%
Late (0.67–1.00)	w/o IRC	weak ($n = 1534$, 49.9%)	66.8% / 0.0% / 33.2%
		base ($n = 2$, 0.1%)	0.0% / 0.0% / 100.0%
		strong ($n = 1536$, 50.0%)	66.7% / 33.3% / 0.0%
	with IRC	weak ($n = 3066$, 99.8%)	0.6% / 97.0% / 2.4%
		base ($n = 0$, 0.0%)	– / – / –
		strong ($n = 6$, 0.2%)	16.7% / 16.7% / 66.7%

Analysis of Router Coordination. Tables 10–12 further illustrate how IRC coordinates CMR and EDR throughout the denoising trajectory. CMR and EDR control distinct but interdependent aspects of the same hierarchical computation path. CMR determines the spatial configuration presented to the subsequent Mamba blocks by controlling feature mixing and compression, whereas EDR determines how deeply information is propagated under that spatial configuration. Consequently, the appropriate Mamba depth depends not only on the denoising timestep, but also on the spatial representation produced by CMR.

Without IRC, jointly training CMR and EDR does not naturally establish this dependency. The coarse-to-fine CMR policy becomes less structured, while EDR frequently exhibits saturated depth decisions that are insensitive to the selected CMR route. For example, during early denoising, EDR selects the deep route in 100% of cases regardless of whether CMR selects weak, base, or strong mixing. With IRC, the conditional EDR distribution instead changes according to the spatial configuration selected by CMR. During the same early interval, weak CMR is predominantly followed by base EDR (74.5%), whereas base CMR is more frequently paired with deep EDR (54.3%). This conditional relationship also evolves across the denoising trajectory: during the mid interval, strong CMR is paired with shallow EDR in 99.4% of cases, while during the late interval, weak CMR is paired with base EDR in 97.0% of cases.

The marginal routing distributions provide a complementary view of this behavior. With IRC, CMR progressively shifts toward weaker spatial processing as denoising proceeds, selecting the weak route almost exclusively at later timesteps, while EDR simultaneously adapts its propagation depth to the resulting spatial context. Thus, IRC does not simply encourage CMR and EDR to make similar routing decisions; rather, it coordinates their distinct roles so that Mamba propagation depth is selected in relation to both the spatial configuration and the current denoising state. This behavior is consistent with the coarse-to-fine motivation of MoH: stronger spatial transformation and context propagation are available when constructing representations from highly noisy inputs, while later stages favor lighter spatial processing as fine-grained information emerges. Consistent with this coordinated behavior, IRC improves FID from 11.00 to 9.51 under the same fixed top-3 SSR setting.

Analysis of Routing Behavior across Timesteps and Stages. Figures 7–9 further visualize how the routing decisions evolve across denoising timesteps and hierarchical stages. As shown in Fig. 7,

CMR exhibits a clear transition in spatial processing: stronger mixing is preferred during earlier timesteps, whereas the weak route becomes dominant as denoising progresses. This suggests that the model allocates more explicit spatial transformation when constructing representations from noisy inputs and progressively reduces such processing as finer structures emerge.

EDR exhibits a complementary, stage-dependent behavior. At the highest-resolution stage (Fig. 9), deep Mamba processing dominates the early timesteps and is gradually replaced by lighter propagation as denoising proceeds. At the lower-resolution stage (Fig. 8), the routing distribution is more diverse across shallow, base, and deep configurations, indicating that the required propagation depth also depends on the spatial scale at which Mamba operates. Thus, EDR does not follow a single depth schedule shared across the hierarchy, but adapts Mamba propagation jointly to the denoising state and representation scale.

Finally, Fig. 6 shows that SSR also changes the active representation levels throughout denoising. The 32×32 stage remains a primary pathway, while the relative use of the 16×16 and 8×8 stages varies substantially with timestep, demonstrating that the model dynamically adjusts its multi-scale representation rather than relying on a fixed hierarchy. Taken together, Figs. 7–9 show that spatial mixing, Mamba propagation depth, and active representation scale all evolve along the denoising trajectory. These learned routing patterns are consistent with the coarse-to-fine nature of diffusion: stronger aggregation and propagation are emphasized when forming structure from noisy representations, while the computation shifts toward finer and lighter processing as denoising progresses.

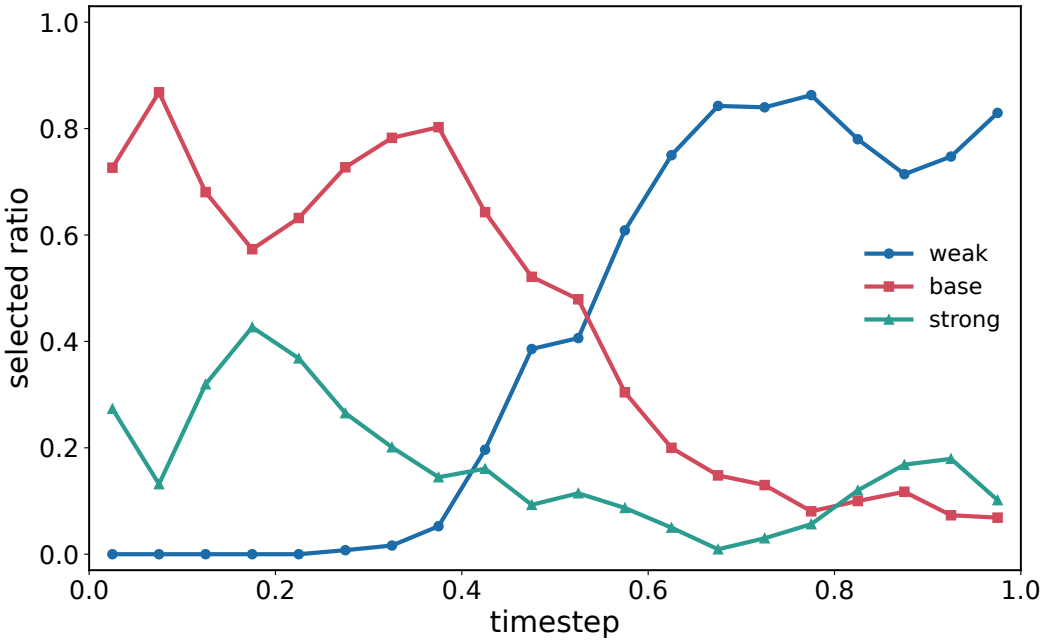


Figure 7: CMR behavior at third stage (8×8 resolution).

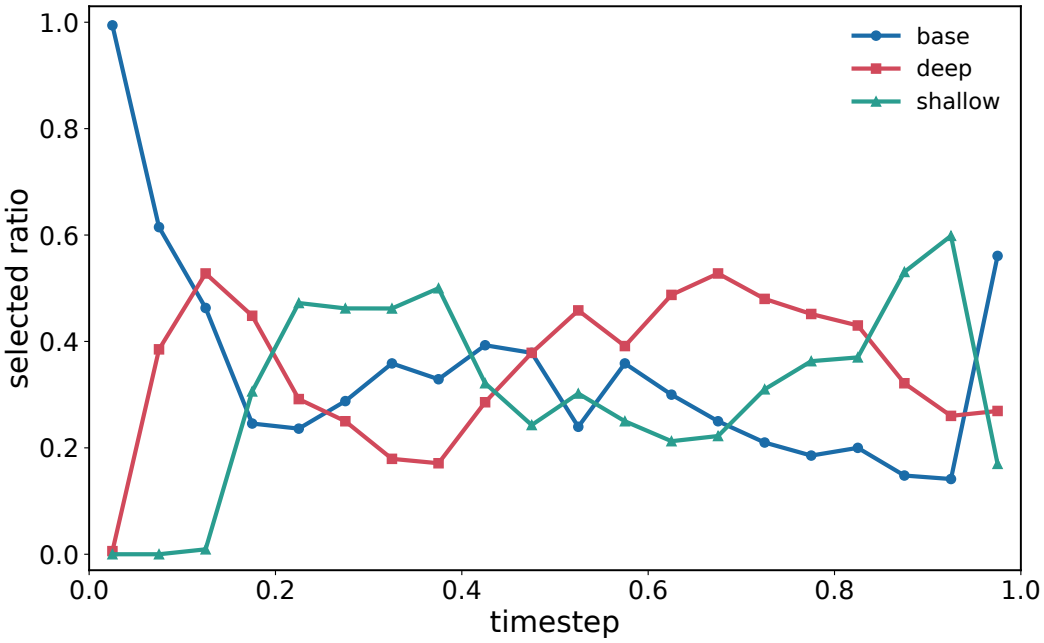


Figure 8: EDR behavior at third stage (8×8 resolution).

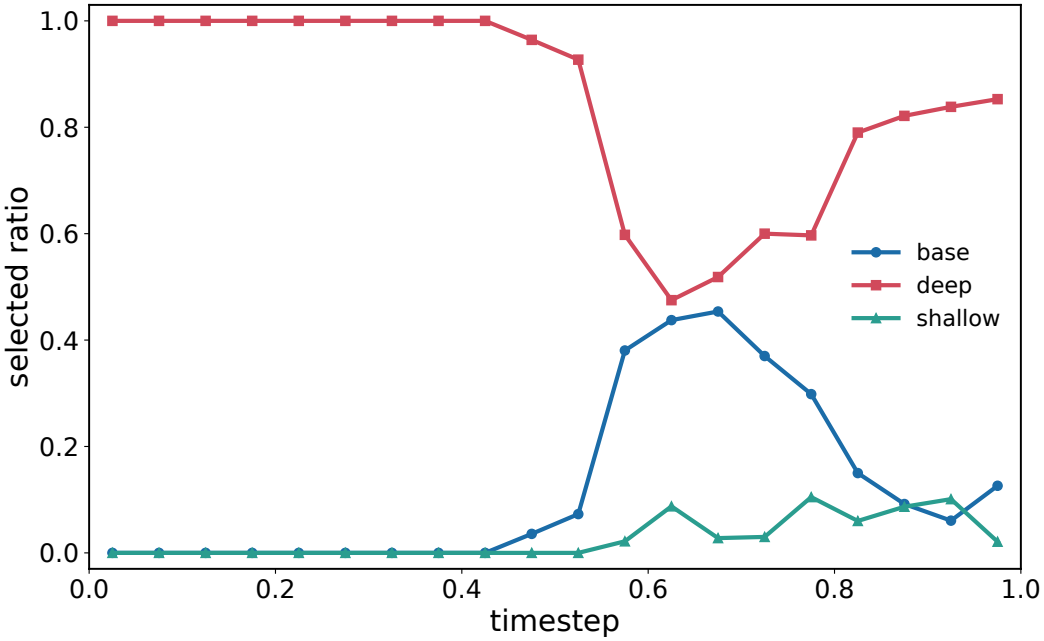


Figure 9: EDR behavior at first stage (32×32 resolution).

A.5 Training Stability

As shown in Fig. 10 and Fig. 11, both training loss and FID improve smoothly throughout training, with no evidence of optimization collapse or late-stage instability. MoH reaches an FID of 7.99 at 100K steps. These results indicate that dynamic routing does not hinder convergence.

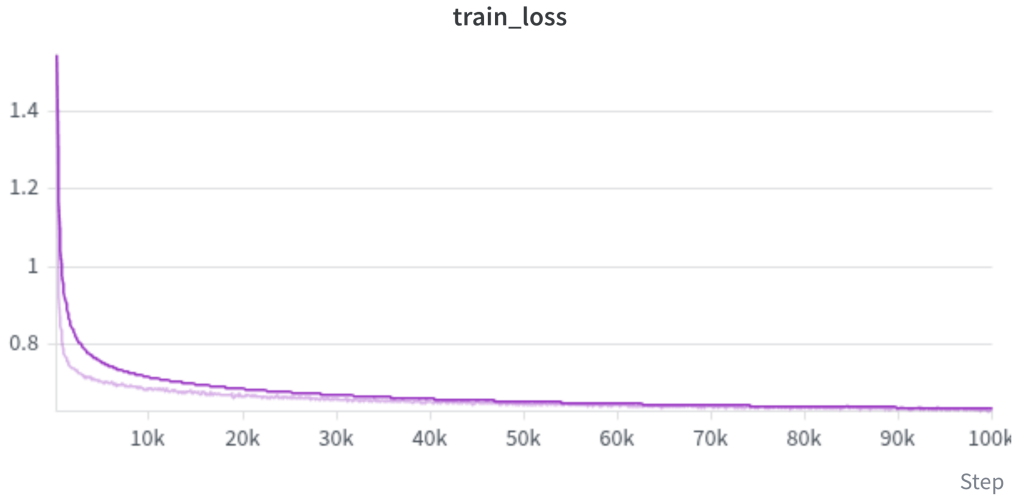


Figure 10: Training loss curves of MoH-B on CelebA-HQ 256×256.

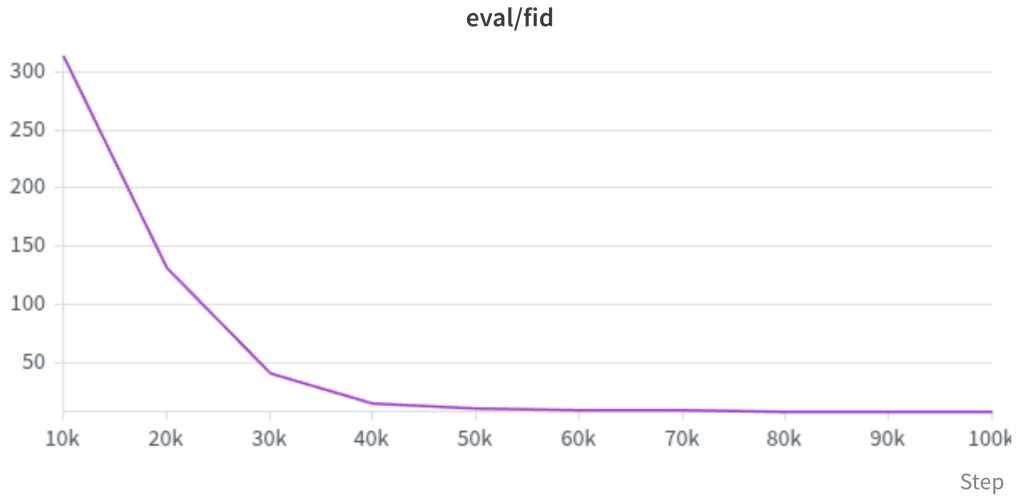


Figure 11: FID^{5k} progression of MoH-B on CelebA-HQ 256×256.

Router Utilization. We further examine whether the learned routers collapse to a small subset of routing choices. As shown in Table 13, all routing options remain actively utilized. SSR selects all three stages with non-trivial frequencies, while CMR and EDR also distribute their decisions across all available compression and depth choices at both hierarchical stages. These results indicate that MoH learns structured routing policies without collapsing to a fixed computation path.

Table 13: Average router utilization of MoH-B on CelebA-HQ 256×256.

Router / Stage	Selection ratio		
SSR	32: 0.776	16: 0.633	8: 0.591
CMR (16×16)	weak: 0.201	base: 0.216	strong: 0.583
CMR (8×8)	weak: 0.261	base: 0.286	strong: 0.453
EDR (32×32)	shallow: 0.027	base: 0.125	deep: 0.848
EDR (16×16)	shallow: 0.195	base: 0.362	deep: 0.443
EDR (8×8)	shallow: 0.324	base: 0.381	deep: 0.295

A.6 Qualitative results

Fig. 12 and Fig. 13 show generated samples from **MoH-B** on CelebA-HQ 256×256 and MS-COCO 256×256. On CelebA-HQ, MoH-B generates visually coherent facial structures with fine-grained details. On MS-COCO, the model produces diverse scenes and object compositions, demonstrating its applicability to more complex image distributions.



Figure 12: Visualization of generated samples on CelebA-HQ 256×256.

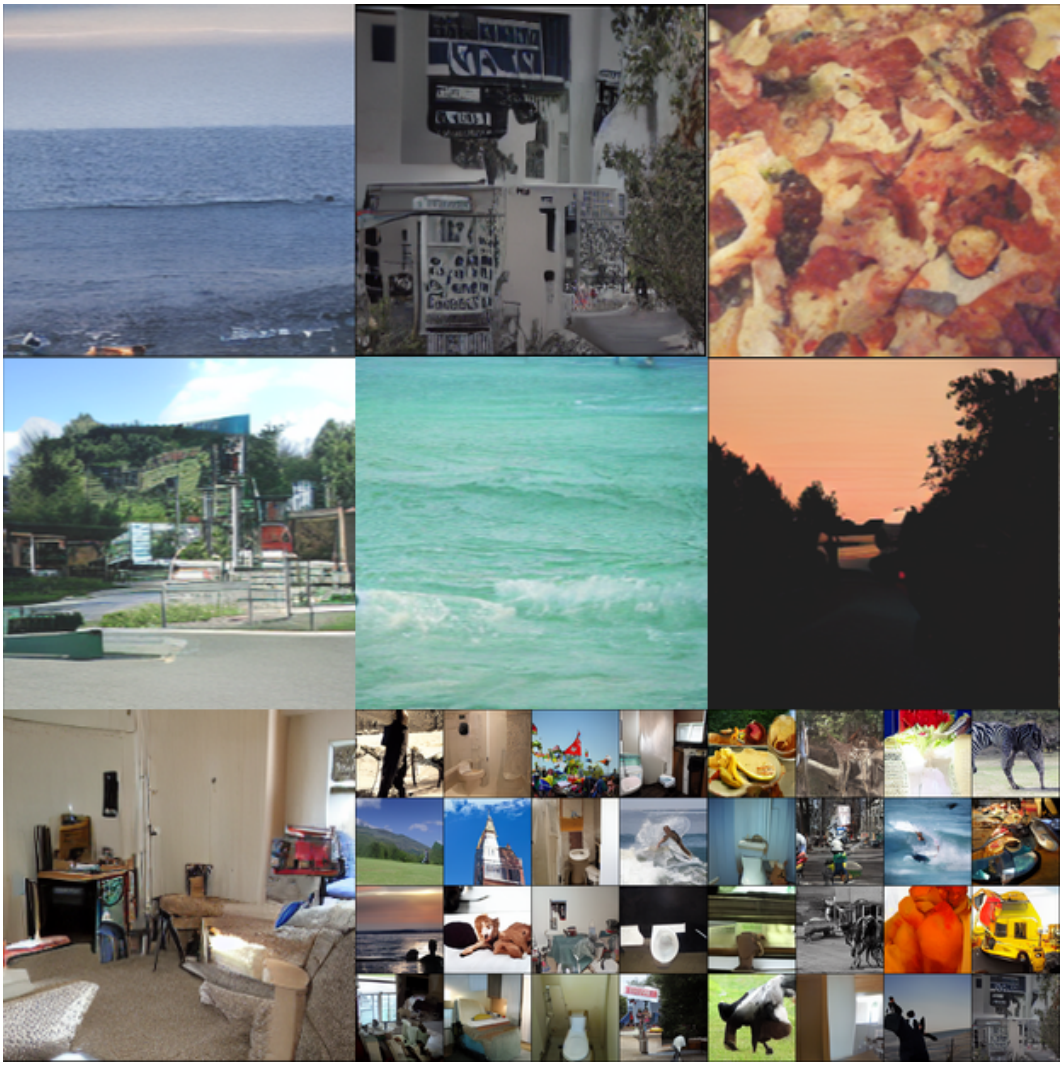


Figure 13: Visualization of generated samples on MS-COCO 256×256.

B Limitations

Due to limited computational resources, our experiments are primarily conducted at 256×256 resolution and do not cover substantially larger model scales or higher-resolution generation. We plan to extend the evaluation to larger-scale and higher-resolution generation in future work.

C License note

CelebA-HQ is used for non-commercial research purposes under the applicable dataset usage terms. MS-COCO annotations are distributed under CC BY 4.0, while the images retain their respective Flickr licenses. We use both datasets for research purposes and follow the applicable usage terms.

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